

# MATH 74: ALGEBRAIC TOPOLOGY

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## 1. TOPOLOGICAL SPACES & HOMEOMORPHISMS

**Definition 1.1.** A **topology** on a set  $X$  is a collection  $J$  of subsets of  $X$  having the following properties.

- (1)  $\emptyset, X \in J$
- (2) The union of the elements of any sub-collection of  $J$  is in  $J$ .
- (3) The intersection of finitely many elements in  $J$  is in  $J$ .

A set for which a topology has been specified is said to be a topological space. We think of it as a pair  $(X, J)$ .

**Definition 1.2.** Elements of  $J$  are called **open sets** of the topological space  $(X, J)$ .

**Definition 1.3.** A subspace  $Y$  of a topological space  $(X, J)$  is a subset  $Y \subset X$  which is given the **subspace topology** where the open sets of  $Y = \{U \cap Y : U \in J\}$

**Definition 1.4.** A space  $X$  is **path connected** if every pair of points of  $X$  can be joined by a path in  $X$ .

**Definition 1.5.** Let  $(X_1, J_1)$  and  $(X_2, J_2)$  be topological spaces. A map  $f : (X_1, J_1) \rightarrow (X_2, J_2)$  is said to be **continuous** if  $f^{-1}(U) \in J_1$  for all  $U \in J_2$ .

**Definition 1.6.** Let  $(X, J)$  and  $(X', J')$  be topological spaces. Then they are **homeomorphic** if  $\exists$  a continuous bijection  $f : (X, J) \rightarrow (X', J')$ . So that  $f^{-1}$  is also continuous.

*Exercise 1.1.* Show that any two open intervals of  $\mathbb{R}$  are homeomorphic (where we put subspace topology on the open intervals).

*Solution.* Let  $(a, b)$  and  $(c, d)$  be open intervals. **TODO:**

**Remark 1.7.** Number of path connected topologies is preserved by continuous maps. For instance,  $[0, 1]$  is not homeomorphic to  $(0, 1)$ , and  $\mathbb{R}$  is not homeomorphic to  $\mathbb{R}^2$ .

*Exercise 1.2.*

- (a) Is  $\mathbb{R}^2$  homeomorphic to  $\mathbb{R}^3$ ?
- (b) Is  $S^2 (= x^2 + y^2 + z^2 = 1)$  homeomorphic to  $T^2 = S^1 \times S^1$ ?
- (c) Is  $S^2$  homeomorphic to  $S^1 \times [a, b]$ ?

*Solution.*

- (a) **TODO:**

**Definition 1.8.** If  $f$  and  $f'$  are continuous maps from spaces  $X$  to  $Y$  we say that  $f$  is homotopic to  $f'$  if  $\exists$  a continuous map  $F : X \times I = [0, 1] \rightarrow Y$  such that  $F(x, 0) = f(x)$  and  $F(x, 1) = f'(x)$ .  $F$  is called the **homotopy** between  $f$  and  $f'$ , and we write  $f \simeq f'$ .

**Definition 1.9.** If  $X$  and  $Y$  are topological spaces, we can define a **product topology** on  $X \times Y$  by saying that a basis consists of the subsets  $U \times V$  as  $U$  ranges over open sets in  $X$  and  $V$  ranges over open sets in  $Y$ .

**Definition 1.10.** Define a **path**  $f$  as  $f : [0, 1] \rightarrow X$  with  $f(0) = x_0$  and  $f(1) = x$ .

**Definition 1.11.** Two paths  $f, f' : [0, 1] \rightarrow X$  are said to be path homotopic if there exist a continuous map  $F : I \times I \rightarrow X$  such that  $\forall s \in I$ ,

$$F(s, 0) = f(s), \quad F(s, 1) = f'(s)$$

and  $\forall t \in I$ ,

$$F(0, t) = x_0, \quad F(1, t) = x$$

Then we write  $f \underset{p}{\simeq} f'$ .

**Example 1.1.** Let  $f : X \rightarrow \mathbb{R}^2$  and  $g : X \rightarrow \mathbb{R}^2$  then  $f \simeq g$ .

*Proof.* Want to find  $F$  such that  $F : X \times I \rightarrow \mathbb{R}^2$ .  $F(x, 0) = f(x)$ ,  $F(x, 1) = g(x)$  for all  $x \in X$ . The following map (straight line homotopy) works.

$$F(x, t) = (1 - t)f(x) + tg(x)$$

□

**Lemma 1.12.** The relations  $\simeq$  and  $\simeq_p$  are equivalence relations.

*Proof.*

- (1) Reflexive:  $f \simeq f$  by  $F(s, t) = f(s)$ .
- (2) Symmetric: Let  $f \simeq g$  by the homotopy  $F(s, t)$ . We can show  $g \simeq f$  by  $F(w, t) = F(s, 1 - t)$ .
- (3) Transitive: Let  $f \simeq g$  by  $F(s_1, t)$  and  $g \simeq h$  by  $G(s_2, t)$ . We can show that  $f \simeq h$  by  $H(s, t) = \begin{cases} s = s_1, & 0 \leq t \leq 1/2 \\ s = s_2, & 1/2 \leq t \leq 1 \end{cases}$  **TODO: check and also prove for  $\simeq_p$**

Thus path-homotopies are equivalence relations and we write  $[f]$  to denote the path-homotopy equivalence class of  $f$ . □

**Lemma 1.13** (Pasting Lemma). Let  $X = A \cup B$  be a topological space with  $A, B \subset X$  are closed. Let  $f : A \rightarrow Y, g : B \rightarrow Y$  be continuous and  $f(x) = g(x), \forall x \in A \cap B$ , then

$$h(x) = \begin{cases} f(x), & x \in A \\ g(x), & x \in B \end{cases}$$

is continuous.

**Definition 1.14.** Let  $f : I \rightarrow X$  be a path from  $x_0$  to  $x_1$  and  $g : I \rightarrow X$  be a path from  $x_1$  to  $x_2$ , then we define a new path

$$f \star g : I \rightarrow X$$

from  $x_0$  to  $x_2$  by

$$f \star g(t) = \begin{cases} f(2t), & t \in [0, 1/2] \\ g(2t - 1), & t \in [1/2, 1] \end{cases}$$

where  $\star$  is **composition of paths**.

Some Properties of ' $\star$ '.

(1) ' $\star$ ' is well-defined operation in the path homotopy equivalence class,

$$[f] \star [g] = [f \star g]$$

(2) Identity:

$$[f] \star [e_{x_1}] = [f], \quad [e_{x_0}] \star [f] = [f]$$

where  $e_{x_t}$  is the constant path based at  $x \in X$  and  $t \in [0, 1]$

(3) Inverse:

$$[f] \star [\bar{f}] = [e_{x_0}], \quad [\bar{f}] \star [f] = [e_{x_1}]$$

where  $\bar{f}(t) = f(1 - t)$ .

(4) Associative:

$$[f] \star ([g] \star [h]) = ([f] \star [g]) \star [h]$$

*Proof.*

(1) Let  $f \simeq_p f'$  by  $F$  and  $g \simeq g'$  by  $G$ , then we can show that  $f \star g \simeq f' \star g'$  by defining  $H : I \times I \rightarrow X$  such that

$$H(s, t) = \begin{cases} F(2s, t), & s \in [0, 1/2] \\ G(2s - 1, t), & s \in [1/2, 1] \end{cases}$$

(2)  $f \star e_{x_1} \simeq f$  by  $H : I \times I \rightarrow X$  such that

$$H(s, t) = \begin{cases} f\left(\frac{2s}{2-t}\right), & s \in [0, \frac{2-t}{2}] \\ x_1, & s \in [\frac{2-t}{2}, 1] \end{cases}$$

(3)  $f \star \bar{f} \simeq_p e_{x_0}$ .

Define a path  $\alpha(t)$  that starts at  $x_0$  and ends at  $f(t)$ . Then the path-homotopy is given by  $\alpha(t) \star \bar{\alpha}(t)$ .

$$H(s, t) = \begin{cases} f(2st), & s \in [0, 1/2] \\ f(2t(1 - s)), & s \in [1/2, 1] \end{cases}$$

**TODO:** straight line homotopy

(4) **TODO:**

□

**Remark 1.15.** Note that  $\star$  operation on the set of path-homotopy equivalence classes of paths in the space  $X$  forms a groupoid structure.

## 2. THE FUNDAMENTAL GROUP

**Definition 2.1.** Let  $X$  be a space and  $x_0 \in X$  be a point in  $X$  we say that a path  $f : I \rightarrow X$  is a **loop** based at  $x_0$  if  $f(0) = f(1) = x_0$

**Definition 2.2.** The set of all path-homotopy equivalence classes of loops based at  $x_0$  forms a group under  $\star$  called the **fundamental group** of  $x$  based at  $x_0$ , denoted as  $\pi_1(X, x_0)$

- Identity:  $[e_{x_0}]$
- Inverses:  $[f] \rightarrow [\bar{f}]$

**Example 2.1.**  $\pi_1(\mathbb{R}^n, (0, 0)) \equiv 0 \equiv \langle e_{origin} \rangle$

*Proof.* Take any two loops  $f(t), g(t)$  based at the origin. Then consider the straight-line homotopy

$$H(s, t) = f(s)(1 - t) + tg(s)$$

**TODO:** (Check that it is a path homotopy) □

**Definition 2.3.** Let  $\alpha$  be a path from  $x_0$  to  $x_1$ , where  $x_0, x_1 \in X$ . We define a map  $\hat{\alpha} : \pi_1(X, x_0) \rightarrow \pi_1(X, x_1)$  such that

$$\hat{\alpha}([f]) := [\bar{\alpha}] \star [f] \star [\alpha] = [\bar{\alpha} \star f \star \alpha]$$

**Proposition 2.4.**  $\hat{\alpha}$  is well-defined and an isomorphism.

*Proof.* Let  $f \simeq_p f'$ . We need to check that

$$\begin{aligned} [\bar{\alpha}] \star [f] \star [\alpha] &= [\bar{\alpha}] \star [f'] \star [\alpha] \\ \iff [\bar{\alpha} \star f \star \alpha] &= [\bar{\alpha} \star f' \star \alpha] \\ \iff \bar{\alpha} \star f \star \alpha &\simeq_p \bar{\alpha} \star f' \star \alpha \end{aligned}$$

Since  $\star$  is well-defined,  $[f] \star [\alpha] = [f'] \star [\alpha]$ ,  $\hat{\alpha}$  is well-defined

Let  $[f], [g] \in \pi_1(X, x_0)$ .

$$\begin{aligned} \hat{\alpha}([f]) \star \hat{\alpha}([g]) &= [\bar{\alpha}] \star [f] \star [\alpha] \star [\bar{\alpha}] \star [g] \star [\alpha] \\ &= [\bar{\alpha}] \star [f] \star [e_{x_0}] \star [g] \star [\alpha] \\ &= [\bar{\alpha}] \star [f] \star [g] \star [\alpha] \\ &= \hat{\alpha}([f] \star [g]) \end{aligned}$$

Hence  $\hat{\alpha}$  is a homomorphism. To check that  $\hat{\alpha}$  is an isomorphism, we see that  $\hat{\hat{\alpha}}(\hat{\alpha}([f])) = [f]$  for all  $[f] \in \pi_1(X, x_0)$ . □

**Corollary 2.4.1.** The fundamental group of  $X$  based at  $x_0$  only depends on the path-connected component of  $x_0 \in X$  upto isomorphism.

**Definition 2.5.** Let  $h : X \rightarrow Y$  be a continuous map such that  $h(x_0) = y_0$ ,  $x_0 \in X, y_0 \in Y$ . Then define  $h_\star : \pi_1(X, x_0) \rightarrow \pi_1(Y, y_0)$  by  $h_\star([f]) = [h \circ f]$ . This is a **push-forward** map.

**Proposition 2.6.**  $h_\star$  is a well-defined homomorphism.

*Proof.* Let  $f \simeq_p f'$ . We need to check that  $h \circ f \simeq_p h \circ f'$ . Let  $H : I \times I \rightarrow X$  be the path homotopy between  $f$  and  $f'$ . Consider  $h \circ H : I \times I \rightarrow Y$ .

$$\begin{aligned} h \circ H(s, 0) &= h \circ f \\ h \circ H(s, 1) &= h \circ f' \\ h \circ H(0, t) &= h \circ f(0) \\ h \circ H(1, t) &= h \circ f(1) = h \circ f(0) \end{aligned}$$

Hence  $h \circ H$  is the required path-homotopy.

Consider  $[f], [g] \in \pi_1(X, x_0)$ . We want to show that  $h_*([f]) \star h_*([g]) = h_*([f] \star [g])$ .

$$h_*([f]) \star h_*([g]) = [h \circ f] \star [h \circ g]$$

Note that it suffices to show that  $h(f \star g) = (h \circ f) \star (h \circ g)$ . Now,

$$\begin{aligned} h(f \star g)(s) &= \begin{cases} (h \circ f)(2s), & s \in [0, \frac{1}{2}] \\ (h \circ g)(2s - 1), & s \in [\frac{1}{2}, 1] \end{cases} \\ &= (h \circ f) \star (h \circ g) \end{aligned}$$

Hence proved. □

**Theorem 2.7.** *If  $h : (X, x_0) \rightarrow (Y, y_0)$  and  $k : (Y, y_0) \rightarrow (Z, z_0)$  then*

- (a)  $(k \circ h)_* = k_* \circ h_*$
- (b) *If  $\text{id} : (X, x_0) \rightarrow (X, x_0)$  is the identity map then  $(\text{id})_*$  is the identity  $\pi_1(X, x_0) \rightarrow \pi_1(X, x_0)$ .*

*Proof.* (a)

$$\begin{aligned} (k \circ h)_*([f]) &= [(k \circ h) \circ f] \\ &= [k(h(f))] \\ &= [k_*([h_*([f])])] \\ &= (k_* \circ h_*)[f] \end{aligned}$$

$$(b) \quad (\text{id})_*[f] = [\text{id} \circ f] = [f]$$

□

**Corollary 2.7.1.** *If  $h : X \rightarrow Y$  is a homeomorphism then  $\pi(X, x_0) \cong \pi(Y, h(x_0))$  via  $h_*$ .*

**Definition 2.8.** *Let  $P : E \rightarrow B$  be a surjective map. A open set  $U \subseteq B$  is said to be **evenly covered** by  $P$  if  $P^{-1}(U) = \bigsqcup_{\alpha} V_{\alpha}$ , where  $V_{\alpha}$  are open sets of  $E$  for all  $\alpha$  and  $P|_{V_{\alpha}}$  is a homeomorphism of  $V_{\alpha}$  onto  $U$ .*

**Definition 2.9.** *Let  $P : E \rightarrow B$  be surjective if every point  $b \in B$  has a neighbourhood  $U \subset B$  such that  $U$  is evenly covered by  $P$ . Then we say that  $P$  is a **covering map** and  $E$  is a **covering space** of  $B$ .*

**Example 2.2.**

(1)  $\text{id} : X \rightarrow X$ .

(2)  $X \times \{1, 2, \dots, n\} \xrightarrow{P} X$  where  $P(X, i) = x$ .

**Theorem 2.10.** The map  $P : \mathbb{R} \rightarrow S'$  such that  $x \mapsto (\cos(2\pi x), \sin(2\pi x))$  is a covering space.

*Proof.*  $P^{-1}(U) = \bigsqcup_{n \in \mathbb{Z}} (n - \frac{1}{4}, n + \frac{1}{4}) =: V_n$

Restricted to each  $V_n$  the map  $P$  is a homeomorphism because  $\sin(2\pi x)$  is strictly monotonic on  $V_n$  for all  $n$ .

**TODO: finish** □

**Definition 2.11.** If a path-connected space  $X$  has  $\pi_1(X, x_0) \cong 0$  then we say that  $X$  is **simply-connected**.

**Lemma 2.12.** If  $X$  is simply-connected, then any two paths in  $X$  with the same initial point and end point are path-homotopic.

*Proof.*  $f \star \bar{g} \simeq_p e_{x_0}$  and  $f \simeq_p f \star e_{x_1} \simeq_p f \star (\bar{g} \star g) \simeq_p (f \star \bar{g}) \star g \simeq_p e_{x_0} \star g \simeq_p g$ . □

**Definition 2.13.** Let  $p : E \rightarrow B$  be a surjective map,  $f : X \rightarrow B$  be a map from  $X \rightarrow B$ . Then a map  $\tilde{f} : X \rightarrow E$  is called **lifting** of  $f$  to  $E$  if  $p \circ \tilde{f} = f$ .

$$\begin{array}{ccc} & E & \\ & \downarrow p & \\ X & \xrightarrow{\tilde{f}} & B \\ & \xrightarrow{f} & \end{array}$$

**Lemma 2.14** (Path Lifting Lemma). Let  $p : E \rightarrow B$  be a covering map with  $p(e_0) = b_0$ . Any path  $f : [0, 1] \rightarrow B$  beginning at  $b_0$  can be lifted uniquely to a path  $\tilde{f} : [0, 1] \rightarrow E$  beginning at  $e_0$ .

*Proof.* Cover  $B$  by open sets  $U$ , each of which is evenly covered by  $p$ . We can find a subdivision (partition) defined by  $s_0, \dots, s_n \in [0, 1]$  such that  $f([s_i, s_{i+1}])$  is contained entirely inside a evenly covered  $U$  for all  $i$ . **TODO: Think about why this is possible.** Now we will define  $\tilde{f}$  inductively. Suppose that  $\tilde{f}(0) = e_0$  and it has been defined for  $0 \leq s \leq s_i$  for some  $i$ . We will now define  $\tilde{f}$  at  $[s_i, s_{i+1}]$ . Suppose that  $f([s_i, s_{i+1}]) \subset U$  where  $U$  is evenly covered open set of  $B$ . This implies that  $p^{-1}(U) = \bigsqcup_{\alpha} V_{\alpha}$ . Now,  $\tilde{f}(s_i)$  lies in one such  $V_{\alpha}$  say  $V_0$ .

Define  $\tilde{f}$  on  $[s_i, s_{i+1}]$  as  $\tilde{f}(s) := (p|_{V_0})^{-1}(f(s))$  for all  $s \in [s_i, s_{i+1}]$ . By pasting lemma  $\tilde{f}$  is continuous on  $[0, s_{i+1}]$  and we argue inductively to define  $\tilde{f}$  for  $[0, 1]$ .

By induction, we can again suppose that for another lift  $\tilde{f}' : [0, 1] \rightarrow E$  of  $f$  with  $\tilde{f}'(0) = e_0$ , it agrees with  $\tilde{f}$  of  $0 \leq s \leq s_i$  for some  $i$

$$\tilde{f}'(s) = \tilde{f}(s), \quad \forall s \in [0, s_i]$$

Now we would like to show that

$$\tilde{f}'(s) = \tilde{f}(s), \quad \forall s \in [s_i, s_{i+1}]$$

$$\tilde{f}'(s) = (p|_{V_0})^{-1} \circ f(s), \quad \forall s \in [s_i, s_{i+1}]$$

$\tilde{f}'(s_i)$  lies in  $V_0$  since  $\tilde{f}'(s_i) = \tilde{f}(s_i) \implies \tilde{f}'([s_i, s_{i+1}])$  lies in  $V_0$ .

$p \circ \tilde{f}' = f$  on  $[s_i, s_{i+1}]$ . But  $\tilde{f}'([s_i, s_{i+1}]) \subset V_0$  and  $p|_{V_0}$  is a homeomorphism.

Then for all  $s \in [s_i, s_{i+1}]$

$$\implies \tilde{f}'(s) = (p|_{V_0})^{-1} \circ f(s) = \tilde{f}(s)$$

□

**Lemma 2.15** (Path Homotopy Lifting Lemma). *Let  $p : E \rightarrow B$  be a covering map. Let  $p(e_0) = b_0$ . Let  $f : I \times I \rightarrow B$  be a map, with  $f(0, 0) = b_0$ . Then there exists a lifting of  $f$  to  $E$  such that  $\tilde{f}(0, 0) = e_0$ . If  $f$  is a path homotopy, so is  $\tilde{f}$ .*

*Proof.* **TODO:**  $I \times I$  is a rectangle. Pick a partition of both sides, then the image of these small rectangles will be contained in  $U$

1. Define  $\tilde{f}$  on  $0 \times I$  and  $I \times 0$  by lifting both of those paths to paths that begin at  $e_0$ .
2. Take a subdivision of  $I \times I$  say  $I_i \times J_j$  such that each of those small rectangles are contained in an evenly covered open set of  $B$ .
3. By induction assume that  $\tilde{f}$  has been defined on  $I_i \times J_j$  such that  $i < i_0, j = j_0$ .
4. Now, to define  $\tilde{f}$  on  $I_{i_0} \times I_{j_0}$  we consider the image of the union of left and bottom edge of  $I_{i_0} \times I_{j_0}$  under  $\tilde{f}$ . This lies inside  $V_0 \subset p^{-1}(U)$ . □

**Remark 2.16.** *The lifted path homomoty is also unique.*

**Lemma 2.17.** *Let  $P : E \rightarrow B$  be a covering map with  $P(e_0) = b_0$ . Let  $f, g$  be two paths in  $B$  from  $b_0$  to  $b_1$ . Let  $\tilde{f}, \tilde{g}$  be their lifts beginning at  $e_0$ . Then if  $f \simeq_p g \implies \tilde{f}$  and  $\tilde{g}$  end at the same point and they are path homotopic.*

*Proof.* Assume  $f \simeq_p g$  with the path homotopy  $F : I \times I \rightarrow B$ . Then Lemma 2.15 implies that  $F$  lifts to  $\tilde{F} : I \times I \rightarrow E$  with  $\tilde{F}(0, t) = e_0$ .

Then we can say that  $\tilde{F}(1, t) = e_1$  for some  $e_1 \in P^{-1}(b_1)$ ,  $\forall t \in I$ . By uniqueness of path lifting,  $f, g$  have unique lifts to  $E$ , beginning at  $e_0$ . We also know that  $\tilde{F}(s, 0)$  and  $\tilde{F}(s, 1)$  are lifts of  $f, g$  both beginning at  $e_0$ .

Then  $\tilde{F}(s, 0) = \tilde{f}$  and  $\tilde{F}(s, 1) = \tilde{g}$  for all  $s \in I$ . Hence  $\tilde{F}(1, 0) = \tilde{F}(1, 1) \implies \tilde{f}(1) = \tilde{g}(1)$ , and  $\tilde{f} \simeq \tilde{g}$ . □

**Theorem 2.18.**  $\pi_1(S^1, (1, 0) = b_0) \cong (\mathbb{Z}, +)$

*Proof.*

$$\begin{array}{ccc} & & \mathbb{R} \\ & \nearrow \tilde{f} & \downarrow \\ I & \xrightarrow{f} & S^1 \end{array}$$

Consider the covering map

$$P : \mathbb{R} \rightarrow S^1, P(x) = (\cos(2\pi x), \sin(2\pi x))$$

Let  $[f] \in \pi_1(S^1, b_0)$ . Let  $\tilde{f}$  be the lift of  $f$  to  $\mathbb{R}$  beginning at 0. Now consider  $\tilde{f}(1)$ . It will be an integer since  $\tilde{f}(1) \in P^{-1}(b_0) = P^{-1}((1, 0))$ . Define

$$\phi : \pi_1(S^1, b_0) \rightarrow \mathbb{Z}, \quad \phi([f]) = f(1)$$

By previous lemma,  $\phi$  is well-defined.

Now we show that  $\phi$  is surjective. Let  $n \in \mathbb{Z}$ . Define  $\tilde{f}$  to be the path in  $\mathbb{R}$  which starts at 0 and ends at  $n$ . Then by definition,  $(P \circ \tilde{f})$  will start at  $(1, 0)$  and end at  $(1, 0)$  where  $\tilde{f}$  is by definition the lift of  $(P \circ \tilde{f})$  beginning at 0. So  $\phi([P \circ \tilde{f}]) = n$ .

To show that  $\phi$  is injective suppose that  $\phi([f]) = \phi([g]) = n$  for some  $[f], [g] \in \pi_1(S^1, b_0)$  we need to show that  $[f] = [g] \in \pi_1(S^1, b_0)$ .

Now let  $\tilde{f}$  and  $\tilde{g}$  be two lifts of  $f$  and  $g$  respectively beginning at 0. Moreover, since  $\mathbb{R}$  is simply connected and  $\tilde{f}(0) = \tilde{g}(0) = 0$ ,  $\tilde{f}(1) = \tilde{g}(1) = n$ ,  $\tilde{f} \simeq_p \tilde{g}$ , then  $P \circ \tilde{f}$  will be the path-homotopy between  $f$  and  $g$ . This implies  $[f] = [g]$ .

Now we want to show that  $\phi$  is a group homomorphism. Let  $[f], [g] \in \pi_1(S^1, b_0)$  such that  $\phi([f]) = n$ ,  $\phi([g]) = m$  for  $n, m \in \mathbb{Z}$ . Now we want to show that  $\phi([f] \star [g]) = \phi([f \star g]) = n + m$ . We need to find a lift of  $f \star g$  beginning at 0 and ending at  $n + m$ .

Let  $\tilde{f}, \tilde{g}$  be lifts of  $f, g$  respectively,  $\tilde{f}(0) = \tilde{g}(0) = (0)$ .

Define a path on  $\mathbb{R}$  by

$$h(s) = \begin{cases} \tilde{f}(2s), & s \in [0, \frac{1}{2}] \\ n + \tilde{g}(2s - 1), & s \in [\frac{1}{2}, 1] \end{cases}$$

Then  $h$  is a path beginning at 0 and ending at  $n + m$ . Now consider,

$$P(h(s)) = \begin{cases} P \circ \tilde{f}(2s), & s \in [0, \frac{1}{2}] \\ P \circ (n + \tilde{g}(2s - 1)) = P \circ \tilde{g}(2s - 1), & s \in [\frac{1}{2}, 1] \end{cases}$$

This implies that  $h(s)$  is a lift of  $f \star g(s) \implies \phi([f \star g]) = h(1) = n + m$  □

**Corollary 2.18.1.** *Fundamental Theorem of Algebra.*

*Proof.* Let  $p(z) = z^n + a_1 z^{n-1} + \dots + a_n$ , where  $a_i \in \mathbb{C}$ .



For the sake of contradiction assume that  $p$  has no roots in  $\mathbb{C}$ . Then for each  $r \in \mathbb{R}^+$  define a function

$$f_r(s) = \frac{\frac{p(re^{2\pi is})}{p(r)}}{\left| \frac{p(re^{2\pi is})}{p(r)} \right|} \in \mathbb{C}$$

Note that  $f_r(s)$  is a loop in  $S^1$  based at  $(1, 0)$  for all  $r$ . ( $f_r(0) = (1, 0) = f_r(1)$  and  $|f_r(s)| = 1, \forall s$ ). Note that as  $r$  varies  $f_r$  gives a path homotopy in  $S^1$  between loops that are based at  $(1, 0)$ . Now, note that  $f_0(s)$  is the trivial loop based at  $(1, 0)$ . This implies,

$$[f_r(s)] = [f_0(s)] = 0 \in \pi_1(S^1, (1, 0))$$

Now fix a large value of  $r$  such that  $r > |a_1| + \dots + |a_n|$  and  $r > 1$ . Now, we have for  $|z| = r$

$$|z^n| = |z| \cdot |z^{n-1}| > (a_1 z^{n-1} + \dots + a_n) |z^{n-1}| \geq |a_1 z^{n-1}| + \dots + |a_n| \geq |a_1 z^{n-1}| + \dots + |a_n|$$

Hence,

$$|z^n| > |a_1 z^{n-1} + \dots + a_n|$$

This implies that the polynomial  $p_t(z) = z^n + t(a_1 z^{n-1} + \dots + a_n)$ ,  $\forall t \in [0, 1]$  has no roots in circle  $|z| = r$ . If we now replace  $p$  by  $p_t$  in the definition of  $f_r(s)$  we get

$$f_r(s, t) = \frac{\frac{p_t(re^{2\pi is})}{p_t(r)}}{\left| \frac{p_t(re^{2\pi is})}{p_t(r)} \right|}$$

At  $t = 1$ ,

$$f_r(s, 1) = f_r(s) \in \pi_1(S^1, (1, 0))$$

At  $t = 0$ ,

$$f_r(s, 0) = e^{2\pi i n s}, \quad \forall s \in [0, 1]$$

So  $[f_r(s, 0)] = n \in \mathbb{Z} \cong \pi_1(S^1, (1, 0))$ . But  $f_r(s, t)$  as we vary  $t \in [0, 1]$  gives a path homotopy between  $f_r(s, 0)$  and  $f_r(s, 1) \implies [f_r(s, 0)] = [f_r(s)] \in \pi_1(S^1, (1, 0)) \implies n = 0$  contradicting that  $p$  is not constant.  $\square$

**Remark 2.19.** In general, we have  $P : (E, e_0) \rightarrow (B, b_0)$  be a covering map. If  $E$  is path-connected then there exists a surjection  $\phi : \pi_1(B, b_0) \rightarrow P^{-1}(b_0)$ . If  $E$  is simply-connected, then  $\phi$  is a bijection.

*Proof.* HW1 iirc  $\square$

**Theorem 2.20.**

$$\pi_1(\mathbb{R}^2 \setminus \{(0, 0)\}, b_0) \cong \mathbb{Z} \cong \pi_1(S^1), \quad b_0 \in S^1.$$

*Proof.* Let  $x_0 \in S^1$ . Then the inclusion map

$$J : (S^1, b_0) \rightarrow (\mathbb{R}^2 \setminus \{(0, 0)\}, b_0)$$

induces an isomorphism

$$J_* : \pi_1(S^1, b_0) \rightarrow \pi_1(\mathbb{R}^2 \setminus \{(0, 0)\}, b_0).$$

Consider

$$r : \mathbb{R}^2 \rightarrow S^1, \quad r(x) = \frac{x}{\|x\|}.$$

Observe that  $r \circ J = \text{id}$ , thus

$$r_* \circ J_* = \text{id}.$$

Thus  $J_*$  is injective.

Choose  $[f] \in \pi_1(\mathbb{R}^2 \setminus \{(0,0)\}, b_0)$ . Recall that,

$$J_* \circ r_*([f])$$

is explicitly  $J \circ r \circ f$ . Let

$$g(s) = \frac{f(s)}{\|f(s)\|}.$$

Next show  $g \simeq_p f$ . Define

$$F : I^2 \rightarrow \mathbb{R}^2 \setminus \{(0,0)\}$$

such that

$$F(s, t) = t \cdot \frac{f(s)}{\|f(s)\|} + (1 - t)f(s).$$

$F$  is the straight line homotopy and does not cross  $0$ .

Since  $f(s) \neq 0$  by definition, suppose

$$t \cdot \frac{1}{\|f(s)\|} + (1 - t) = 0$$

implies  $\|f(s)\|$  negative or  $t \notin [0, 1]$ .

Thus  $F$  is well-defined, since  $f \simeq_p g$ ,

$$J_* \circ r_*([f]) = [f].$$

Thus surjection. □

### 3. RETRACTIONS

**Definition 3.1.** If  $A$  is a subspace of  $X$  s.t.  $\exists$  a map

$$r : X \rightarrow A$$

with  $r(a) = a$  for all  $a \in A$ , then we say  $A$  is a retract of  $X$  and  $r$  is a retraction map from  $X$  to  $A$ .

**Definition 3.2.** Let  $A$  be a subspace of  $X$ .  $A$  is a deformation retract of  $X$  if there exists a homotopy

$$H : X \times I \rightarrow X$$

such that

$$\begin{aligned} H(x, 0) &= x \quad \forall x \in X, \\ H(x, 1) &\in A, \\ H(a, t) &= a. \end{aligned}$$

**Theorem 3.3.** Let  $A$  be a deformation retract of  $X$ . Then  $j : (A, x_0) \hookrightarrow (X, x_0)$  induces an isomorphism in  $\pi_1(A, x_0)$  to  $\pi_1(X, x_0)$ . If  $A$  is a retract of  $X$ , then  $j$  is injective.

*Proof.* **TODO: hw**

□

**Remark 3.4.** If  $X$  deformation retracts to  $A$ , then  $\pi_1(X) \cong \pi_1(A)$ .

**Remark 3.5.** Let  $\iota : A \hookrightarrow X$  then  $\iota_* : \pi_1(A) \rightarrow \pi_1(X)$  is injective if  $X$  only retracts to  $A$ .  
**TODO: only retracts??**

#### 4. APPLICATIONS OF $\pi_1$ -COMPUTATIONS

**Theorem 4.1** (Brouwer-Fixed Point Theorem). *Every map  $h : D^2 \rightarrow D^2$  has a fixed point, i.e.  $h(x) = x$  for at least one  $x \in D^2$ .*

*Proof.* Suppose towards contradiction  $h(x) \neq x$  for all  $x \in D^2$ . Define a function  $\gamma : D^2 \rightarrow S^1$  as follows: consider the straight line starting at  $h(x)$  and going to  $x$ . Define  $r(x)$  as the intersection of that line with  $\partial D^2 = S^1$ . Now  $r$  is a continuous function, and  $r|_{S^1} = \text{Id}_{S^1}$ . That is,  $r$  is a retraction of  $D^2$  to  $S^1$ .

Since for the inclusion  $\iota : S^1 \hookrightarrow D^2$  we should have  $\pi_1(S^1, (1, 0)) \xrightarrow{\iota_*} \pi_1(D^2, (1, 0))$  is an injection which gives a contradiction.  $\square$

**Theorem 4.2** (Borsuk-Ulam Theorem). *Every map  $f : S^2 \rightarrow \mathbb{R}^2$  has a pair of anti-podal points ( $x$  is an antipode of  $-x$ ,  $x \in S^2$ ) in  $S^2$  such that  $f(x) = f(-x)$ .*

*Proof.* Suppose that this is false. Then  $f(x) \neq f(-x)$  for all  $x \in S^2$ . We define  $g : S^2 \rightarrow S^1$  by

$$g(x) = \frac{f(x) - (-x)}{\|f(x) - f(-x)\|}$$

Define a loop  $\eta$  by the equator that lies in the  $X - Y$  plane of  $S^2$ .

$$\eta(s) = (\cos(2\pi s), \sin(2\pi s), 0), s \in [0, 1]$$

and let  $h : I \rightarrow S^1$  defined by  $h := g \circ \eta$ . Note that since  $g(x) = -g(-x)$  we have  $h(s + \frac{1}{2}) = -h(s)$  for all  $s \in [0, \frac{1}{2}]$ .

$$\begin{aligned} g(\cos(2\pi(s + 1/2)), \sin(2\pi(s + 1/2)), 0) &= g(-\cos(2\pi s), -\sin(2\pi s), 0) \\ &= -g(\cos(2\pi s), \sin(2\pi s), 0) \end{aligned}$$

By path-lifting lemma, we can lift  $h$  to a path in  $\mathbb{R}$ , say  $\tilde{h}$  starting at 0 such that

$$\tilde{h}\left(s + \frac{1}{2}\right) = \tilde{h}(s) + \frac{q(s)}{2}$$

for some odd integer values  $q(s)$ . But now,  $q(s) = 2(\tilde{h}(s + \frac{1}{2}) - \tilde{h}(s))$  and hence it is also a continuous function. This implies  $q(s)$  is a constant map (since the domain of  $q$  is connected), i.e.  $\tilde{h}(s + \frac{1}{2}) = \tilde{h}(s) + \frac{q}{2}$ .

$$\tilde{h}(1) = \tilde{h}\left(\frac{1}{2}\right) + \frac{q}{2} = \tilde{h}(0) + q$$

This implies  $\tilde{h}(1) = q$  and hence  $[h] \in \pi_1(S^1, (1, 0))$  is not homotopic to the identity. But since  $\eta$  is nullhomotopic in  $S^2$ ,  $g \circ \eta$  is homotopic to a constant map  $[h] = 0 \in \pi_1(S^1, (1, 0))$ . A contradiction!  $\square$

## 5. HOMOTOPY EQUIVALENCES

**Theorem 5.1.**  $\pi_1(X \times Y, x_0 \times y_0) \cong \pi_1(X, x_0) \times \pi_1(Y, y_0)$  for any pair of topological spaces  $X$  and  $Y$ .

*Proof.* Let  $P_1 : X \times Y \rightarrow X$  and  $P_2 : X \times Y \rightarrow Y$  be the respective projections.

$$\phi : \pi_1(X \times Y, x_0 \times y_0) \rightarrow \pi_1(X, x_0) \times \pi_1(Y, y_0)$$

where

$$\phi([f]) = (P_1)_*[f] \times (P_2)_*[f]$$

$\phi$  is surjective: Let  $[g] \in \pi_1(X, x_0)$  and  $[h] \in \pi_1(Y, y_0)$ . Then define  $[g] \times [h] : I \rightarrow X \times Y$  by  $s \mapsto g(s) \times h(s)$ . Then  $\phi([g] \times [h]) = [g] \times [h]$ .

$\phi$  is injective: Suppose that  $[f] \in \pi_1(X \times Y, x_0 \times y_0)$  such that  $\phi([f]) = (P_1)_*([f]) \times (P_2)_*([f]) = 0$ . Then  $P_1 \circ f \simeq_p e_{x_0}$  and  $P_2 \circ f \simeq_p e_{y_0}$  by homotopies  $F_t$  and  $G_t$ .

Then,  $F_t \times G_t : I \rightarrow X \times Y$  gives  $f \simeq_p e_{x_0 \times y_0} \implies [f] = 0$ .

$$\implies \pi_1(S^1 \times S^1) \cong \mathbb{Z} \times \mathbb{Z}. \quad \square$$

**Definition 5.2.** We say  $f : X \rightarrow Y$  is a homotopy equivalence if there exists a  $g : Y \rightarrow X$  such that  $f \circ g \simeq \text{id}_Y$  and  $g \circ f \simeq \text{id}_X$ . Then we say that  $X$  is homotopy equivalent to  $Y$  and write it as  $X \simeq_h Y$ .

**Remark 5.3.** Homotopy equivalence is an equivalence relation on topological spaces.

**Proposition 5.4.** Suppose that  $X$  deformation retracts to  $A$ . Then  $X \simeq_h A$ .

*Proof.* Let  $H_t : X \rightarrow A$  be the deformation retraction. Then consider  $H_1(X) \subseteq A$ , and  $H_1|_A = \text{id}$ . Let us call  $r := H_1$  and let  $\iota : A \hookrightarrow X$  be the inclusion map. Then  $r \simeq \text{id} \implies \iota \circ r \simeq \text{id}$  and  $r \circ \iota = \text{id} \implies X \simeq_h A$ .  $\square$

**Remark 5.5.** Note not all homotopy equivalence come from deformation retractions.

**Remark 5.6.**  $\pi_1$  is not only an invariant of homeomorphism class of a topological space but also an invariant of the homotopy equivalence class of the space.

**Theorem 5.7.** Let  $\phi : X \rightarrow Y$  be a homotopy equivalence. Then  $\phi_* : \pi_1(X, x_0) \rightarrow \pi_1(Y, \phi(x_0))$  is an isomorphism.

**Lemma 5.8.** If  $\phi_t : X \rightarrow Y$  be a homotopy and  $h$  is the path given by  $\phi_t(x_0)$  for a choice of  $x_0 \in X$ , then we have  $(\phi_0)_* = \widehat{h}((\phi_1)_*)$  on  $\pi_1(X, x_0)$ . That is,

$$\begin{array}{ccc} \pi_1(X, x_0) & \xrightarrow{(\phi_1)_*} & \pi_1(Y, \phi_1(x_0)) \\ & \searrow (\phi_0)_* & \downarrow \widehat{h} \\ & & \pi_1(Y, \phi_0(x_0)) \end{array}$$

*Proof.* Let  $[f] \in \pi_1(X, x_0)$ . Let  $h_t$  be a reparameterization of the path  $h$  such that  $h_t(s) := h(st)$ . In particular,  $h_t$  is a path from  $\phi_0(x_0)$  to  $\phi_t(x_0)$ . Now the composition,  $h_t \star \phi_t \circ f \star \bar{h}_t$  gives a path homotopy between

$$h_0 \star \phi_0 \circ f \star \bar{h}_0 = e_{x_0} \star \phi_0 \circ f \star \bar{e}_{x_0} \simeq \phi_0 \circ f$$

and

$$h_1 \star \phi_1 \circ f \star \bar{h}_1 = h \star \phi_0 \circ f \star \bar{h} \implies (\phi_0)_\star = \widehat{h}((\phi_1)_\star)$$

□

**Theorem 5.9.** *Let  $\phi : X \rightarrow Y$  be a homotopy equivalence. Then  $\phi_\star : \pi_1(X, x_0) \rightarrow \pi_1(Y, \phi(x_0))$  is an isomorphism.*

*Proof.* Since  $\phi$  is a homotopic equivalence, there exists a  $\psi : Y \rightarrow X$  such that  $\phi \circ \psi \simeq \text{id}_Y$  and  $\psi \circ \phi \simeq \text{id}_X$ . Consider,

$$\pi_1(X, x_0) \xrightarrow{\phi_\star} \pi_1(Y, \phi(x_0)) \xrightarrow{\psi_\star} \pi_1(X, \psi(\phi(x_0))) \xrightarrow{\phi_\star} \pi_1(Y, \phi(\psi(\phi(x_0))))$$

Note that  $\psi_\star \circ \phi_\star = (\psi \circ \phi)_\star = \widehat{h} \circ (\text{id}_X)_\star$  for some path  $h$  in  $X$  (by the previous lemma). This implies  $\psi_\star \circ \phi_\star = \widehat{h} \implies \psi_\star$  is surjective since  $\widehat{h}$  is an isomorphism.

$\psi_\star$  is injective because

$$\phi_\star \circ \psi_\star = \widehat{h'} \circ (\text{id}_Y)_\star$$

for some path  $h'$  in  $Y$  (by the previous lemma).

Hence,  $\phi_\star = (\psi_\star)^{-1} \circ \widehat{h}$  which is an isomorphism.

□

## 6. $\pi_1(S_n)$

**Lemma 6.1.** *If a space  $X$  is the union of a collection of path-connected open sets  $A_\alpha$  each containing the base point  $x_0 \in X$ . Let  $A_\alpha \cap A_\beta$  be path connected for any  $\alpha, \beta$ , then every loop in  $X$  based at  $x_0$  is path homotopic to a composition of loops each contained in a single  $A_\alpha$  for some  $\alpha$  and based at  $x_0$ .*

*Proof.* Let  $[f] \in \pi_1(X, x_0)$ . Suppose that  $0 = S_0 < S_1 < \dots < S_m = 1$  is a subdivision of  $[0, 1]$  such that  $f|_{[S_{i-1}, S_i]}$  is contained in a single  $A_\alpha$  for all  $i$  and some  $\alpha$  depending on  $i$ . Let us denote that  $A_\alpha$  by  $A_i$ .

Note  $f(0) = x_0$ . Define  $g_i$  to be the path from  $x_0$  to  $f(S_i) \in A_i \cap A_{i+1}$ .

Consider,

$$(f \star \overline{g_1}) \star (g_1 \star f \star \overline{g_2}) \star (g_2 \star f \star \overline{g_3}) \star \dots \star (g_{m-1} \star f) \simeq_p f$$

while each of them are loops based at  $x_0$  in  $A_i$ s as  $i$  varies. □

**Theorem 6.2.**  $\pi_1(S^n, x_0) \cong \{0\}$  if  $n \geq 2$ .

*Proof.* On  $S^2$  consider two open sets  $A_1 = S^2 \setminus \{\text{north pole}\}$  and  $A_2 = S^2 \setminus \{\text{south pole}\}$ . Let  $[f] \in \pi_1(S^2, x_0)$ . Then by the lemma  $f \simeq_p g_1 \star g_2$  where  $[g_1] \in \pi_1(A_1, x_0)$  and  $[g_2] \in \pi_1(A_2, x_0)$  for some  $g_1$  and  $g_2$  loops based at  $x_0$ . Then,

$$[f] = 0 \in \pi_1(S^2, x_0) \implies \pi_1(S^2, x_0) = 0$$

□

**Remark 6.3** (Stereographic Projection).  $S^n \setminus \{\text{some pt}\}$  is homeomorphic to  $\mathbb{R}^n$ .

## 7. COVERING SPACES (AGAIN)

**Definition 7.1.** **TODO:**  $\mathbb{RP}^n := S^n / \{x \sim -x\}$  (quotient space)